



THE SURAT TECHNICAL EDUCATION & RESEARCH SOCIETY

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Smt. Ushaben Jayvadan Bodawala BCA College &

M.Sc. (IT) Integrated College & B.Sc. Data Science College

Affiliated To Veer Narmad South Gujarat University

ONLINE SHOES WEBSITE (StepUp Footwear)

Subject: 505 Advance Web Designing (AWD)

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Overview

Project Definition: A modern, full-stack footwear e-commerce Single Page Application (SPA) built using the MERN stack for BCA Semester 5.

Footwear Domain Focus: Dedicated online shoe store offering top athletic and casual footwear brands such as Nike, Adidas, Puma, Reebok, Skechers, and Converse.

Footwear Sizing Solution: Solves the primary footwear shopping challenge by providing explicit UK size selection (UK 6 to 11) with dynamic stock validation.

MERN Architecture: Combines React 18 & Vite frontend with Node.js/Express REST APIs and MongoDB document database.

Dual Operation Modules: Customer Storefront (catalog search, category filters, size selection, CartContext, Razorpay payment) and Admin Back-Office (product CRUD, orders status, reviews).

Introduction

Traditional Footwear Retail Issues: Shoppers travel to crowded physical shoe stores, spend hours searching racks, and often face disappointment when their specific shoe size is out of stock.

Manual Store Overhead: Physical footwear retailers struggle with paper stock registers, manual bill calculations, and delayed customer feedback handling.

Proposed MERN Solution: StepUp Footwear offers centralized real-time catalog visibility, instant shoe size availability, and automated subtotal/discount calculations.

Single Page Application (SPA): Built using React 18 and react-router-dom (v6), enabling instantaneous page transitions without full browser reloads.

Academic Relevance: Demonstrates mastery in Advance Web Designing (AWD) concepts: reusable React components, Context API state management, RESTful routing, and NoSQL modeling.

Scope & Objectives

PROJECT SCOPE

- Customer authentication with security CAPTCHA and JWT token storage.
- Footwear catalog with real-time brand filtering and category classification.
- Shoe details page featuring interactive UK size selection (Sizes 6 to 11).
- Global client state management using React Context (AuthContext & CartContext).
- Seamless order checkout with shipping address validation and Razorpay modal.
- Customer star ratings (1-5 stars), written reviews, and Contact Us messaging.
- Comprehensive Admin back-office for managing shoes, categories, brands, and orders.

PROJECT OBJECTIVES

- To build a responsive, high-performance footwear shopping SPA using React and Vite.
- To eliminate size-ordering confusion by requiring customers to choose UK shoe sizes.
- To implement secure RESTful APIs using Express.js, JWT tokens, and Bcrypt hashing.
- To model flexible, document-oriented footwear collections in MongoDB with Mongoose.
- To integrate Razorpay Standard Checkout in test mode for realistic payment simulation.
- To apply advanced modern web development practices taught in BCA Semester 5.

Environment & Technology Stack

HARDWARE & RUNTIME ENVIRONMENT

Development Machine

Intel Core i3 / AMD Ryzen 3 or higher, minimum 4 GB RAM (8 GB recommended for concurrent Node & Vite), 500 MB storage.

Operating System

Microsoft Windows 10 / Windows 11 (64-bit).

Server Runtime

Node.js (v18+) asynchronous event-driven JavaScript runtime environment.

Database Engine

MongoDB (awd_shoes_db) NoSQL document database with Mongoose ODM.

Development Tools

Visual Studio Code, MongoDB Compass, Postman REST API client.

SOFTWARE & WEB FRAMEWORKS (MERN)

Frontend Framework

React 18 with Vite build tool for ultra-fast Hot Module Replacement (HMR).

Client Routing & State

react-router-dom v6 for client routing; React Context API (AuthContext, CartContext).

Backend Framework

Express.js RESTful web framework with CORS, Multer, and JWT authentication.

Payment Gateway

Razorpay Standard Checkout Modal (Test Mode sandbox integration).

Data Dictionary (Part 1)

Database: awd_shoes_db — Collections: users, products, categories

Collection	Field Name	Data Type	Constraint	Description
users	_id	ObjectId	Primary Key	Unique MongoDB identifier for customer account
users	name	String	Required	Full name of customer
users	email	String	Required, Unique	Customer email address for login and notifications
users	password	String	Required	Bcrypt hashed password string
users	role	String	Dflt 'customer'	User role: 'customer' or 'admin'
users	address	String	Optional	Saved delivery street address
products	_id	ObjectId	Primary Key	Unique identifier for shoe model
products	name	String	Required	Footwear title (e.g. Air Zoom Pegasus, Ultraboost)
products	brand	String	Required	Brand manufacturer (Nike, Adidas, Puma, etc.)
products	category	ObjectId	FK (categories)	Reference to footwear category
products	price	Number	Required	Shoe price in Indian Rupees (INR)
products	stock	Number	Required	Available stock quantity in inventory
products	sizes	Array of Num	Required	Supported UK sizes array: [6, 7, 8, 9, 10, 11]
categories	_id	ObjectId	Primary Key	Unique category identifier
categories	name	String	Required, Unique	Category name: Running, Casual, Formal, Boots

Data Dictionary (Part 2)

Database: awd_shoes_db — Collections: orders, reviews, brands, contactmessages

Collection	Field Name	Data Type	Constraint	Description
orders	_id	ObjectId	Primary Key	Unique MongoDB order identifier
orders	user	ObjectId	FK (users)	Customer account placing order
orders	items	Array of Docs	Required	Subdocuments containing product ID, UK size, quantity
orders	totalAmount	Number	Required	Final calculated invoice total in INR
orders	shippingAddress	String	Required	Customer delivery street and city address
orders	paymentStatus	String	Dflt 'Pending'	Payment status: Pending / Paid
orders	orderStatus	String	Dflt 'Pending'	Status: Pending, Processing, Shipped, Delivered
reviews	_id	ObjectId	Primary Key	Unique review identifier
reviews	product	ObjectId	FK (products)	Reference to reviewed footwear product
reviews	user	ObjectId	FK (users)	Reference to reviewing customer
reviews	rating	Number	Required (1 to 5)	Customer star rating for comfort and quality
brands	_id	ObjectId	Primary Key	Unique brand partner record
brands	name	String	Required, Unique	Footwear brand name (Nike, Adidas, Puma, etc.)
contactmessages	_id	ObjectId	Primary Key	Customer inquiry message identifier

ER Diagram

Entity-Relationship (E-R) Diagram — Online Shoes Website (StepUp Footwear)

Database: awd_shoes_db (MongoDB Document Collections & Mongoose Schemas)

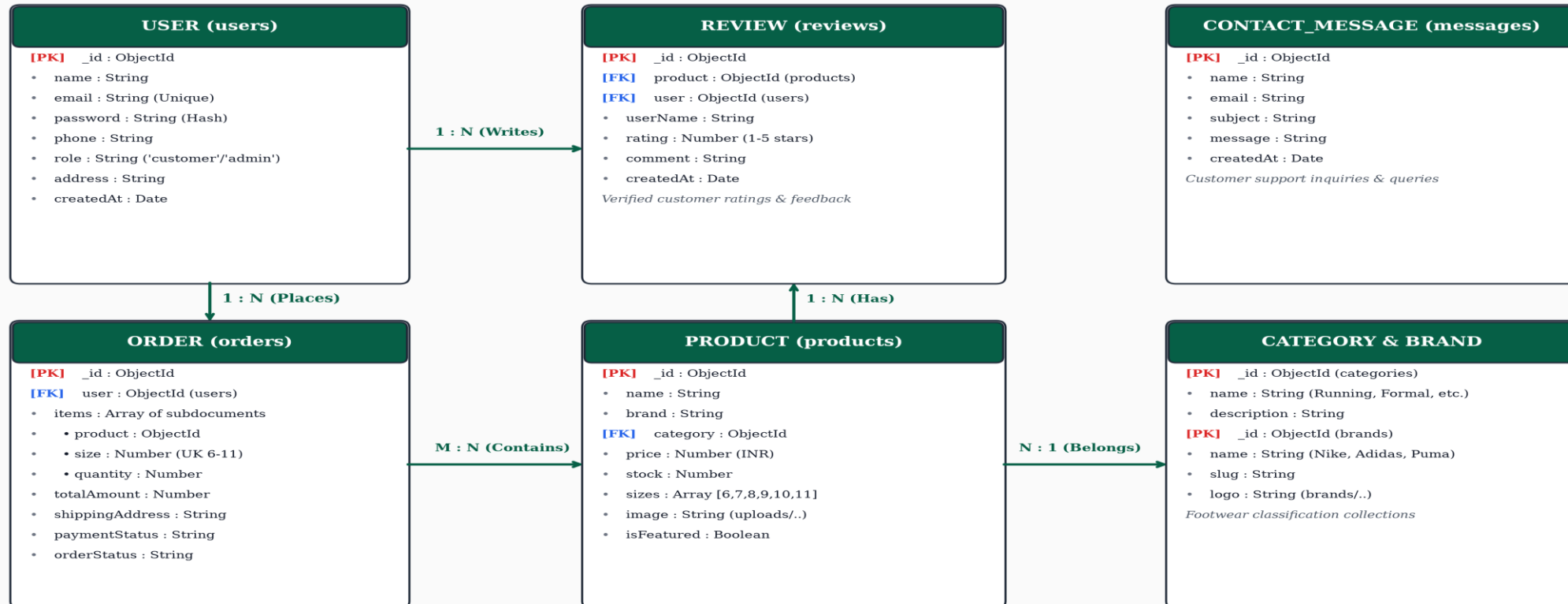


Figure: Entity-Relationship Diagram for Online Shoes Website (awd_shoes_db Collections)

Data Flow Diagram

Data Flow Diagram (DFD Level 0 & Level 1) — Online Shoes Website

Client-Server Flows across React SPA, Node/Express REST APIs, and MongoDB

LEVEL 0: CONTEXT DIAGRAM



LEVEL 1: FUNCTIONAL PROCESSES & DATA STORES

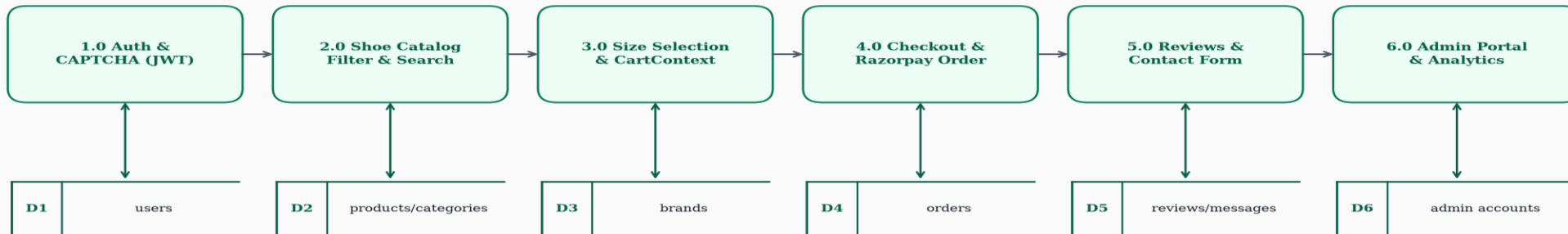


Figure: DFD Level 0 (Context Diagram) and Level 1 Functional Data Flows across React SPA, Node REST APIs, and MongoDB

User Interface: Home Page & Footwear Catalog

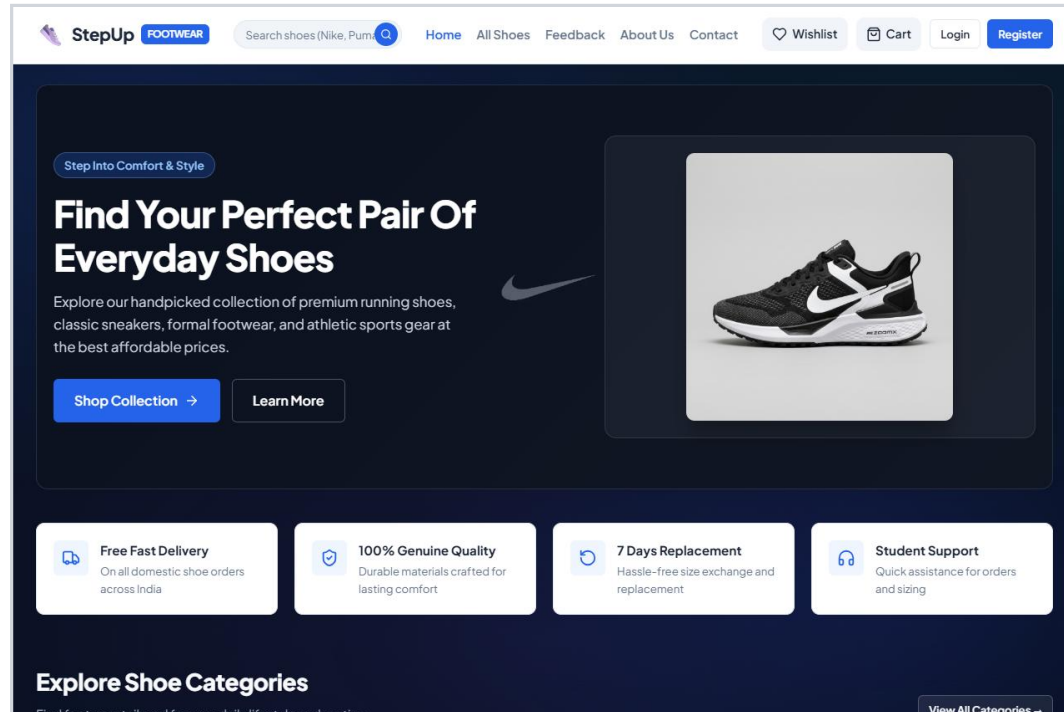


Figure 1: Home Page

Displays modern hero banner, popular brand logos carousel, featured shoes grid, and value highlights.

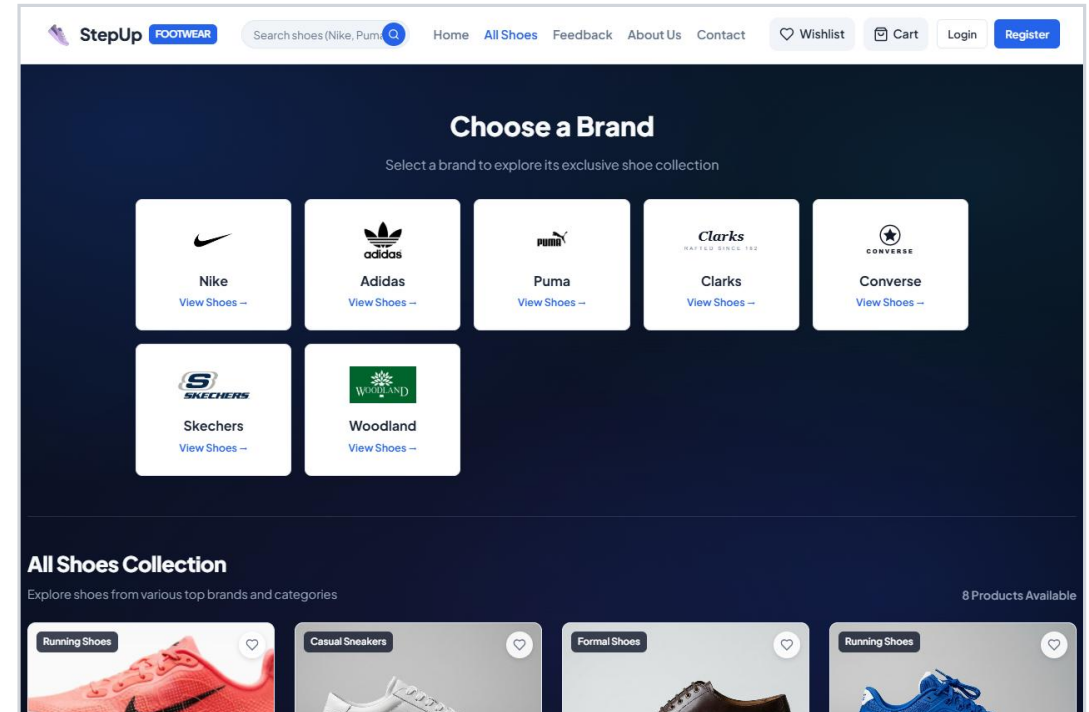


Figure 2: All Shoes Catalog Page

Displays entire footwear collection with brand filters (Nike, Adidas), category badges, and price sorting.

User Interface: Shoe Details & Size Selection

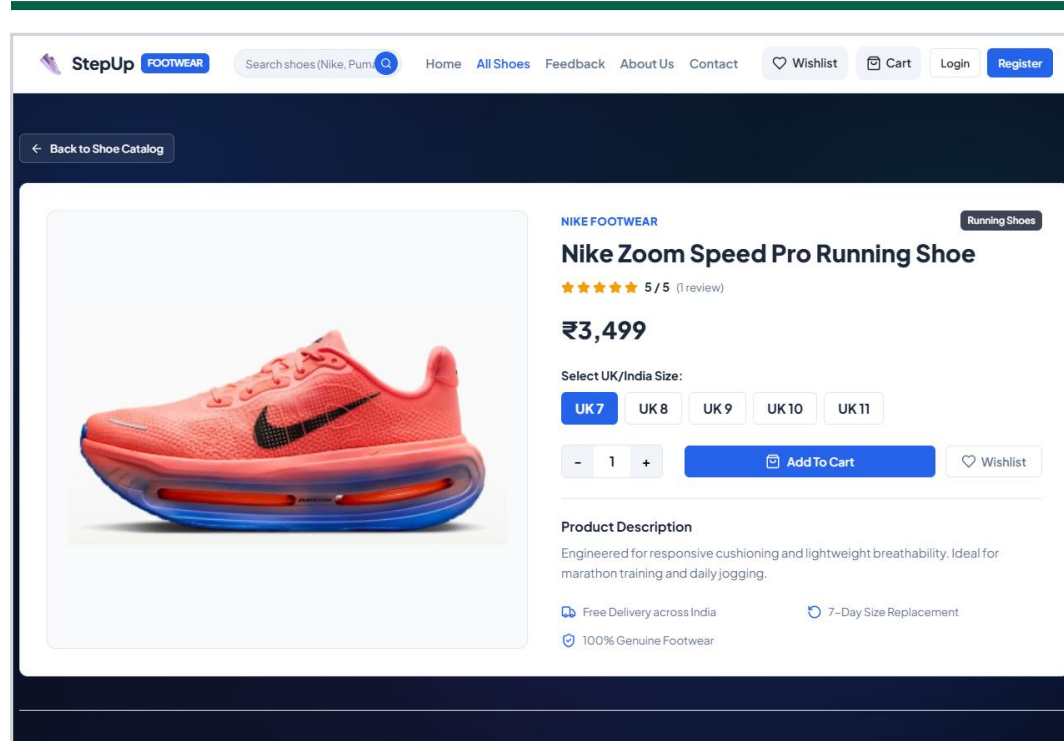


Figure 3: Product Details & UK Sizing

Shows high-resolution shoe imagery, detailed descriptions, and interactive UK size buttons (6 to 11).

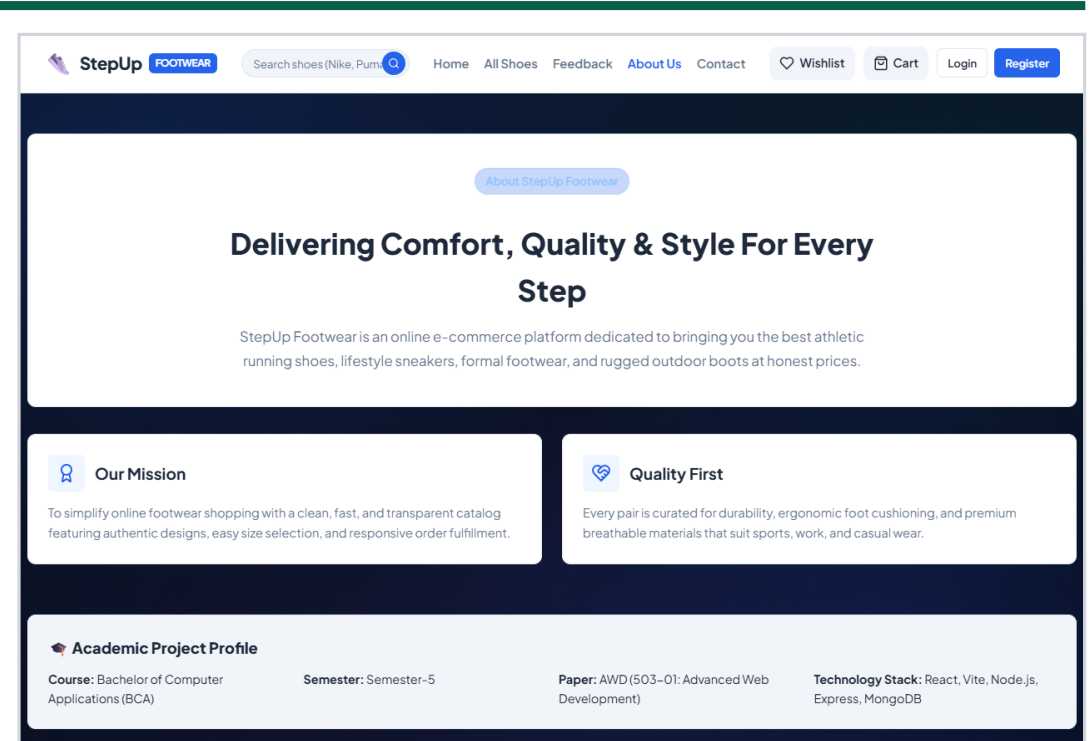


Figure 4: About Us Store Page

Presents store heritage, quality footwear assurance, brand partnerships, and customer support standards.

User Interface: Shopping Cart & Checkout Form

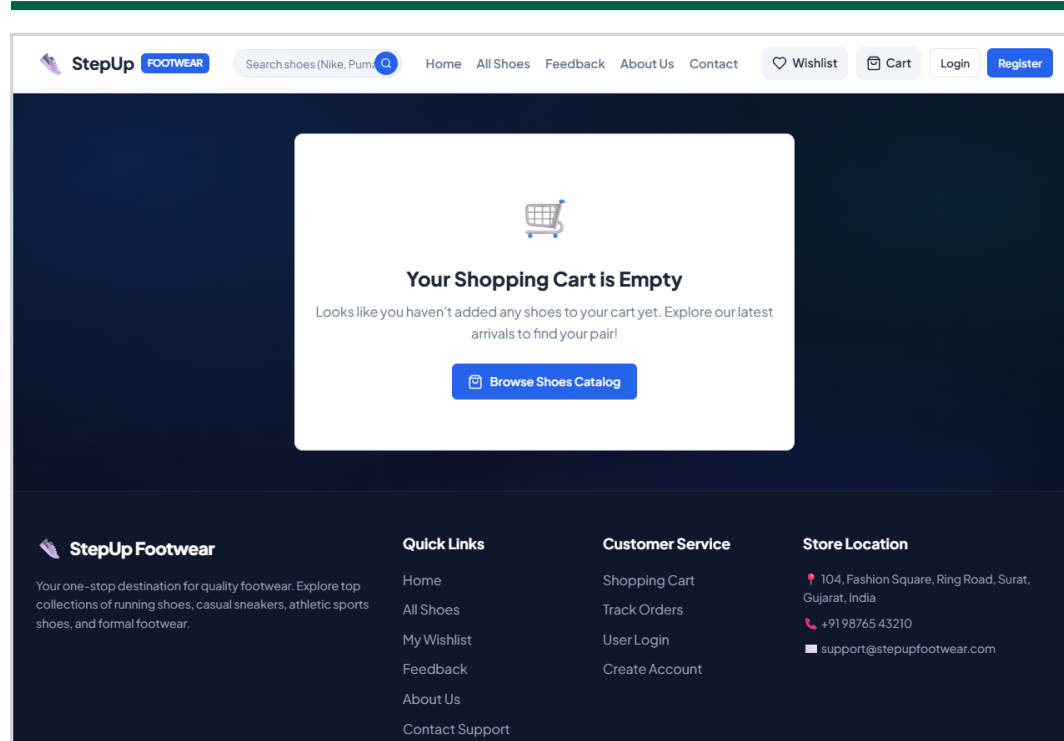


Figure 5: Responsive Shopping Cart

Displays chosen shoes with selected UK size badge, quantity controls, price subtotal, and tax summary.

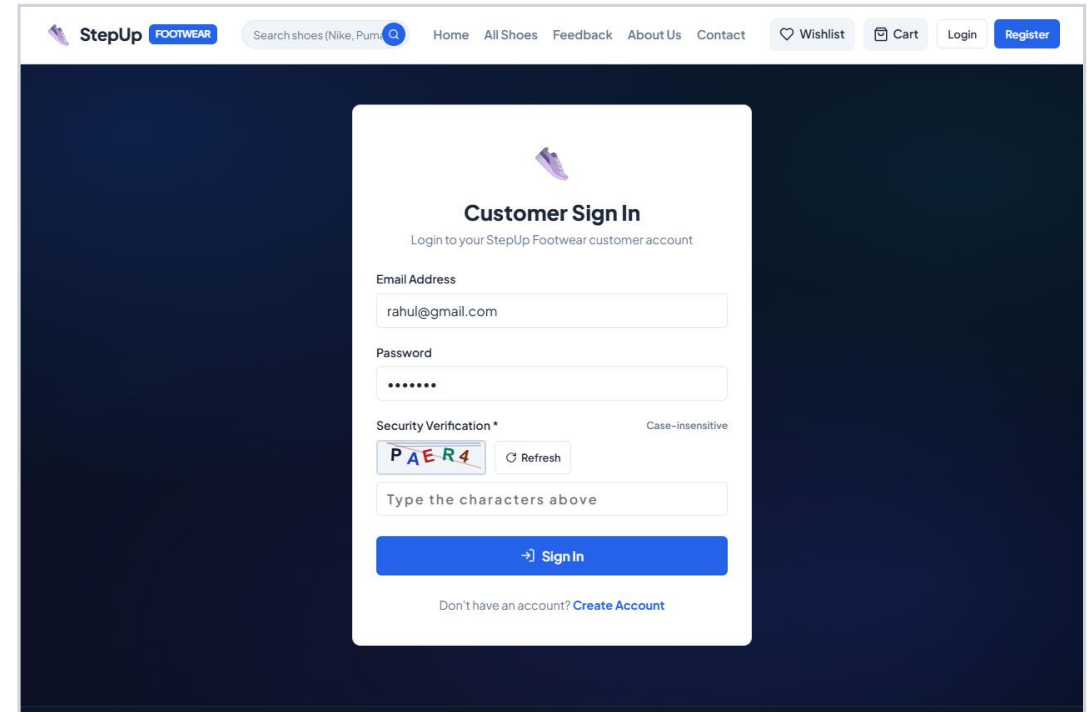
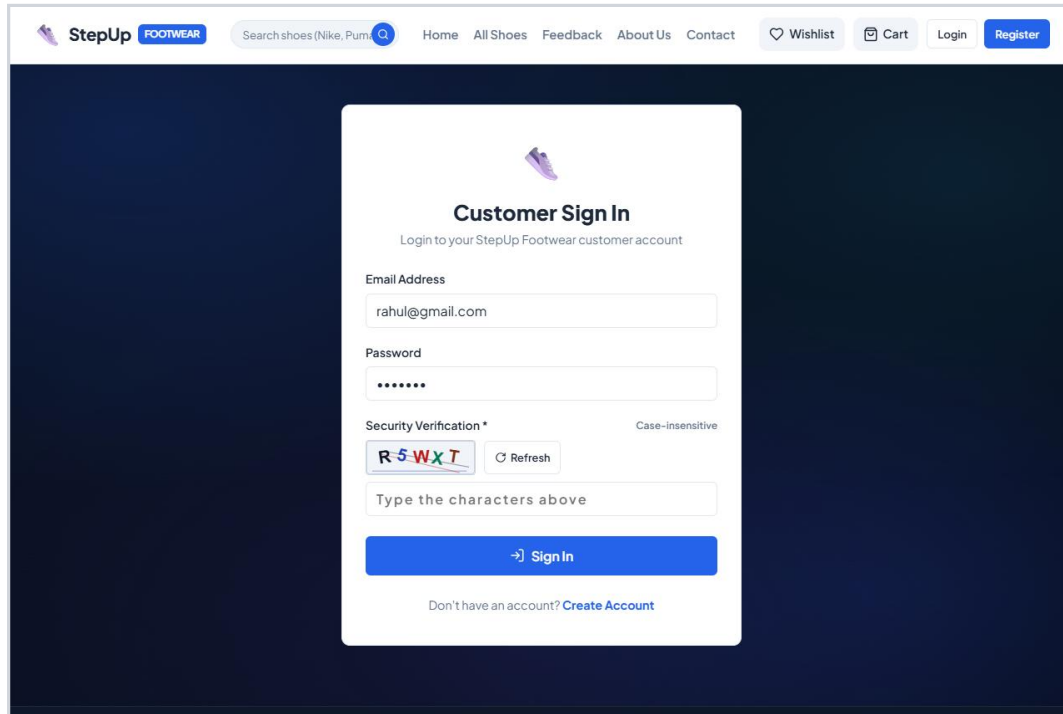


Figure 6: Delivery Checkout & Payment

Collects customer shipping address and triggers the official Razorpay test checkout modal.

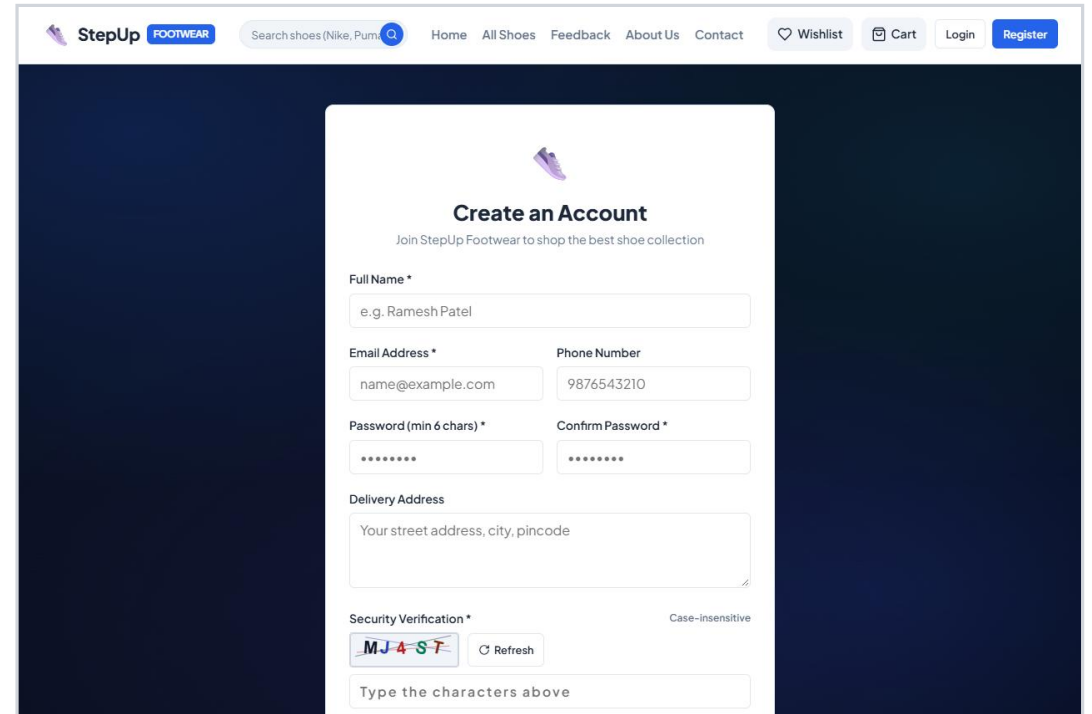
User Interface: Customer Login & Registration



The image shows a web browser window with the 'StepUp FOOTWEAR' logo in the top left. The navigation bar includes a search bar with 'Search shoes (Nike, Puma)', and links for Home, All Shoes, Feedback, About Us, Contact, Wishlist, Cart, Login, and Register. The main content area features a 'Customer Sign In' form. The form has a title 'Customer Sign In' with a shoe icon, a subtitle 'Login to your StepUp Footwear customer account', and a shoe icon. It contains fields for 'Email Address' (with 'rahul@gmail.com' entered), 'Password' (with masked characters), and 'Security Verification *' (with a CAPTCHA image showing 'R-5-W-X-T' and a 'Refresh' button). Below the CAPTCHA is a text input field 'Type the characters above'. A blue 'Sign In' button is at the bottom, with a link 'Don't have an account? Create Account' below it.

Figure 7: Customer Login with CAPTCHA

Secure login form featuring mathematical security CAPTCHA to prevent automated bot login attempts.



The image shows a web browser window with the 'StepUp FOOTWEAR' logo in the top left. The navigation bar includes a search bar with 'Search shoes (Nike, Puma)', and links for Home, All Shoes, Feedback, About Us, Contact, Wishlist, Cart, Login, and Register. The main content area features a 'Create an Account' form. The form has a title 'Create an Account' with a shoe icon, a subtitle 'Join StepUp Footwear to shop the best shoe collection', and a shoe icon. It contains fields for 'Full Name *' (with 'e.g. Ramesh Patel' entered), 'Email Address *' (with 'name@example.com' entered), 'Phone Number' (with '9876543210' entered), 'Password (min 6 chars) *' (with masked characters), 'Confirm Password *' (with masked characters), 'Delivery Address' (with 'Your street address, city, pincode' entered), and 'Security Verification *' (with a CAPTCHA image showing 'M-J-A-S-T' and a 'Refresh' button). Below the CAPTCHA is a text input field 'Type the characters above'.

Figure 8: Customer Registration Form

Customer sign-up interface collecting name, email, phone, password, and CAPTCHA validation.

User Interface: Admin Dashboard & Product Controls

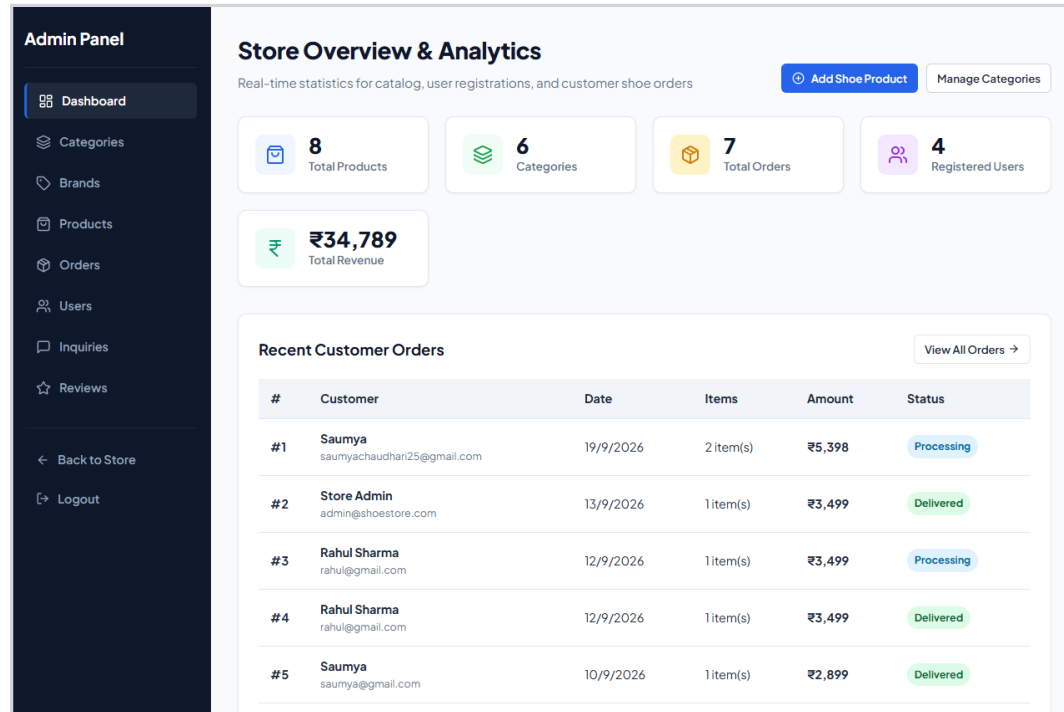


Figure 9: Administrative Dashboard

Overview screen presenting total sales metrics, order counts, product inventory summary, and quick links.

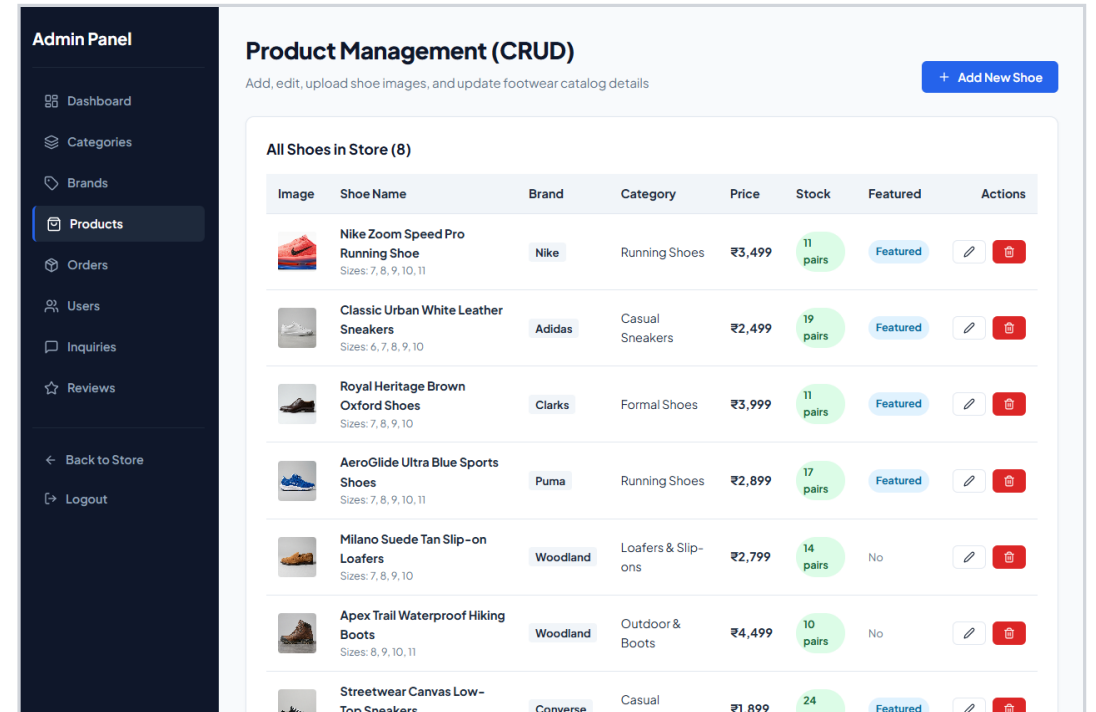


Figure 10: Shoe Products Inventory Table

Allows store managers to add new footwear models, adjust pricing, upload images, and control stock availability.

User Interface: Order Management & Review Moderation

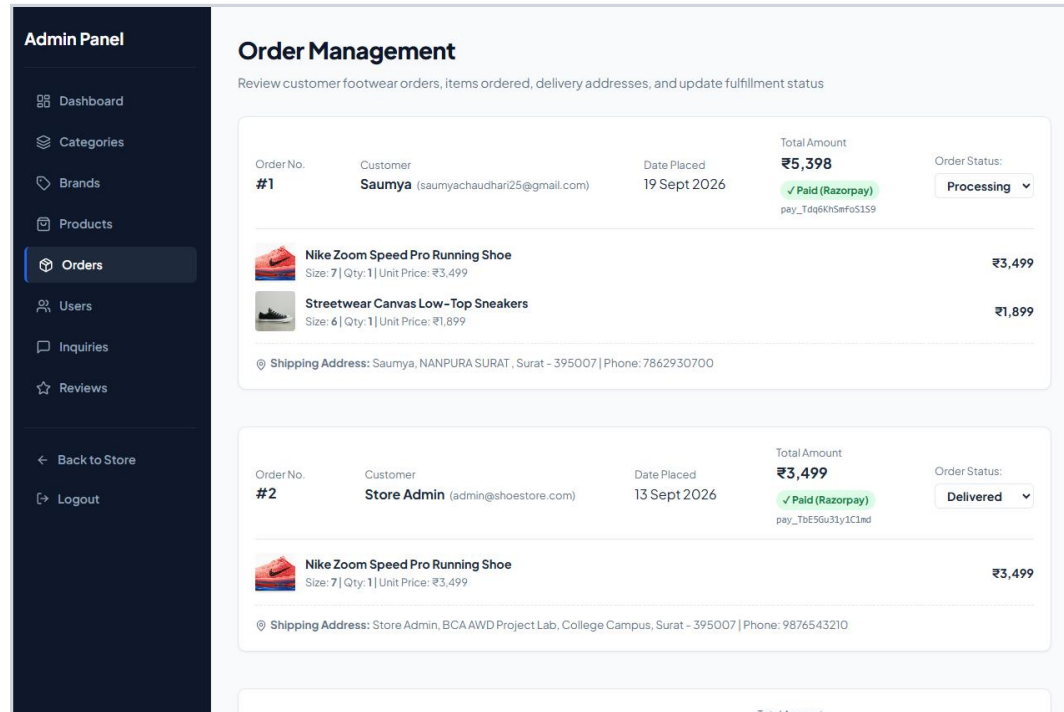


Figure 11: Admin Orders Management

Displays all customer orders, delivery addresses, payment status, and order status update dropdowns.

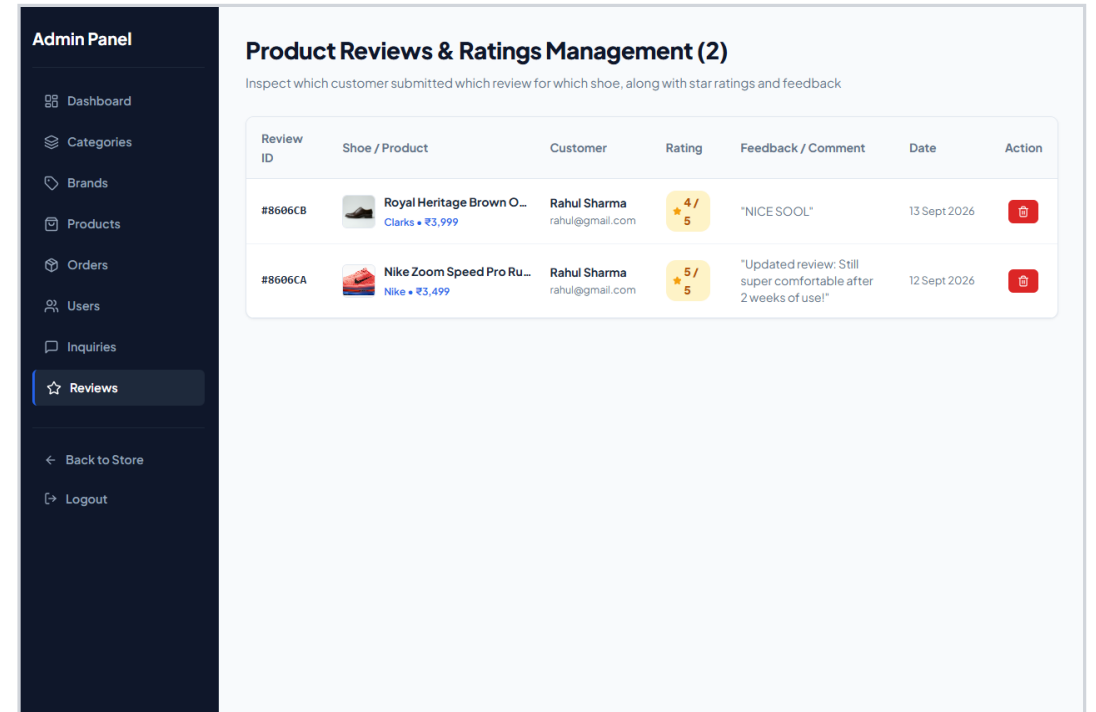


Figure 12: Customer Review Moderation

Enables administrators to view star ratings, monitor customer feedback, and moderate published shoe reviews.

Limitation & System Enhancements

SYSTEM LIMITATIONS

Single Currency Operations

Transactions and pricing are strictly denominated in Indian Rupees (INR — ₹); international currency conversion is not supported.

Single Vendor Architecture

Designed for a single shoe store business; multi-vendor merchant onboarding is not currently supported.

Email Notification Scope

Order receipts are displayed in the app and stored in MongoDB; automated SMTP / Nodemailer email alerts are pending.

Standard Sizing Scope

Offers standard UK shoe sizes (6 to 11) without interactive camera-based foot 3D scanning tools.

FUTURE SYSTEM ENHANCEMENTS

Automated Email & SMS Dispatch

Integrating Nodemailer and SMS APIs (such as Twilio) to send automatic dispatch tracking numbers and delivery alerts.

Interactive Foot Size Advisor

Introducing a foot length measurement calculator to guide customers to their ideal size and reduce return rates.

Customer Return & Refund Portal

Adding a dedicated customer portal for return requests with administrator approval workflows.

Promotional Coupon Code Engine

Enabling administrators to create seasonal discount coupon promo codes with expiry dates.

Progressive Web App (PWA)

Configuring service workers for offline catalog caching and installable mobile app experience on smartphones.

Conclusion

Successful MERN Stack Deployment: The Online Shoes Website (StepUp Footwear) successfully demonstrates a modern, scalable full-stack e-commerce Single Page Application.

Effective Sizing Problem Solution: Effectively addressed the primary problem in online footwear retail through explicit UK size selection integrated into the CartContext.

Robust Backend Architecture: Demonstrated RESTful API design with Express.js, JWT authentication, Bcrypt security, and NoSQL document modeling in MongoDB.

Intuitive User Experience: Achieved instantaneous client-side navigation, interactive filters, CAPTCHA security, and simulated Razorpay payment.

Academic Competency: Provided hands-on mastery of modern web architecture and Advance Web Designing (AWD) principles taught in BCA Semester 5.

References

Textbooks & Academic References:

- Alex Banks and Eve Porcello, 'Learning React: Modern Patterns for Developing React Apps', 2nd Edition, O'Reilly Media, 2020.
- Greg Lim, 'Beginning Node.js, Express & MongoDB Development', Independently Published, 2021.
- Robin Nixon, 'Learning PHP, MySQL & JavaScript: With jQuery, CSS & HTML5', 6th Edition, O'Reilly Media, 2021.
- Roger S. Pressman, 'Software Engineering: A Practitioner's Approach', 8th Edition, McGraw-Hill Education, 2014.

Official Documentation & Online Resources:

- React.js Official Documentation: <https://react.dev/> — Components, hooks (useState, useEffect, useContext), and JSX syntax.
- Node.js & Express.js Documentation: <https://nodejs.org/> and <https://expressjs.com/> — Server runtime and REST APIs.
- MongoDB & Mongoose Documentation: <https://www.mongodb.com/docs/> and <https://mongoosejs.com/docs/> — NoSQL schemas.
- Razorpay Payments Documentation: <https://razorpay.com/docs/> — Standard checkout modal integration.
- MDN Web Docs (Mozilla Developer Network): <https://developer.mozilla.org/> — Modern HTML5, CSS3, and JavaScript standards.

THANK YOU